

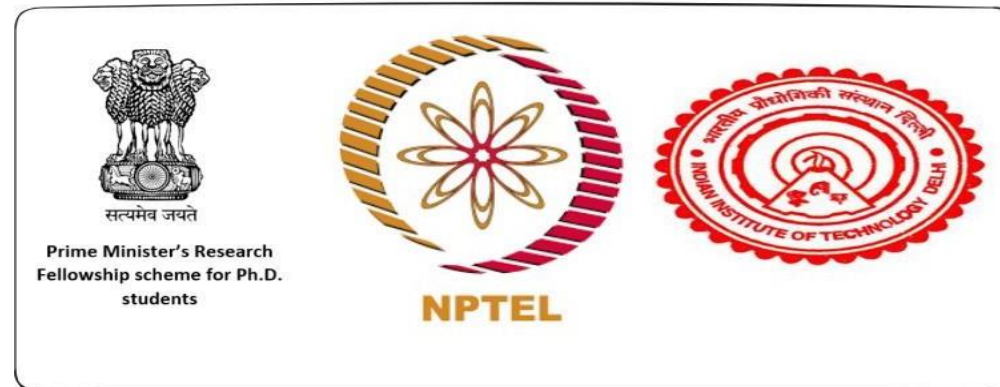
Interaction Session 1

on

Cloud Computing
noc25-cs11

by

Mainak Chakraborty



28th January 2025

Indian Institute of Technology, Delhi

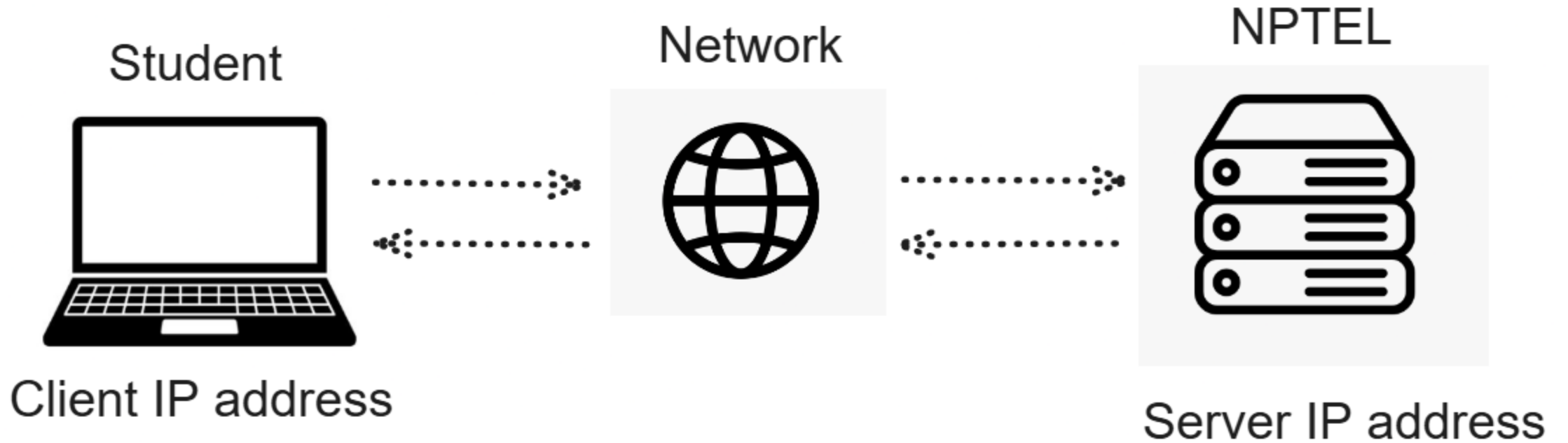
Points to Remember:

- Total 12 Discussion sessions on taught lectures and Tutorials
- Any specific problem, please upload in the discussion forum, I will try to reply within 24 hours
- PPT and Video will be shared with everyone
- Assignments will not be solved but similar problems will be solved.

TOPICS TO BE COVERED

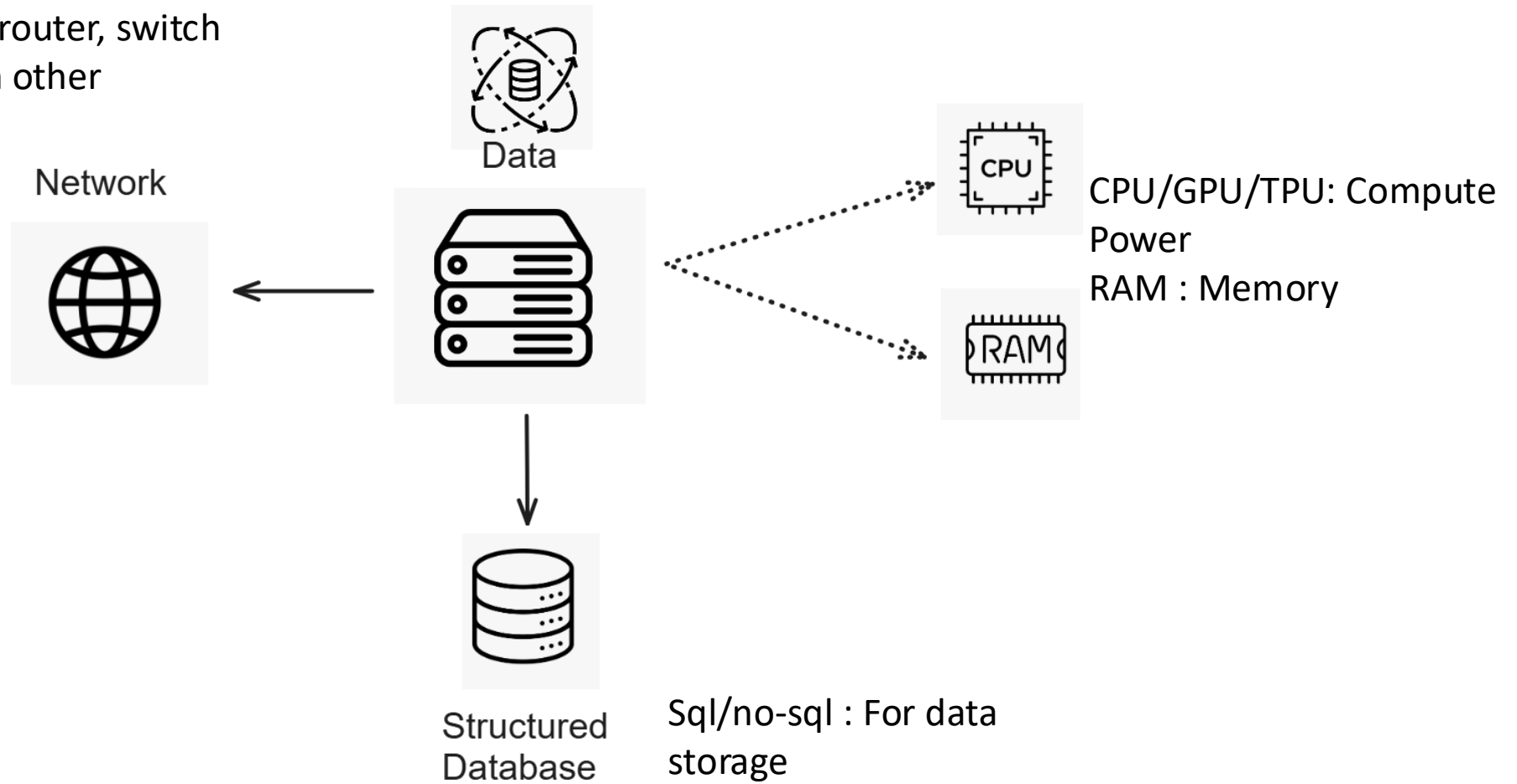
1. Introduction to Cloud Computing
2. Characteristics of Cloud Computing
3. The Deployment Models of the Cloud
4. Types of Cloud Computing
5. Hands-on Session with AWS

Background

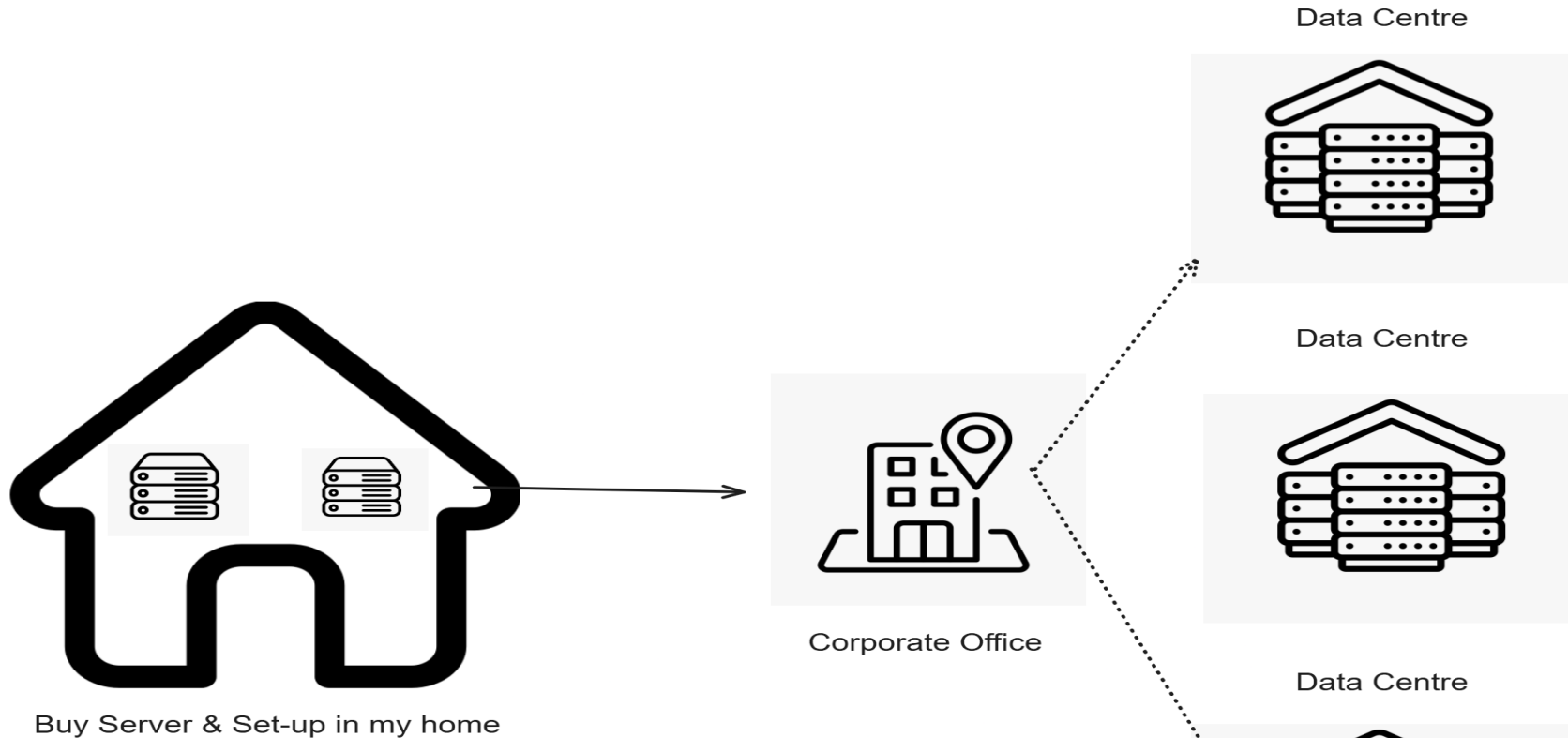


What is the server consisting of?

Network : Cables, router, switch connected to each other



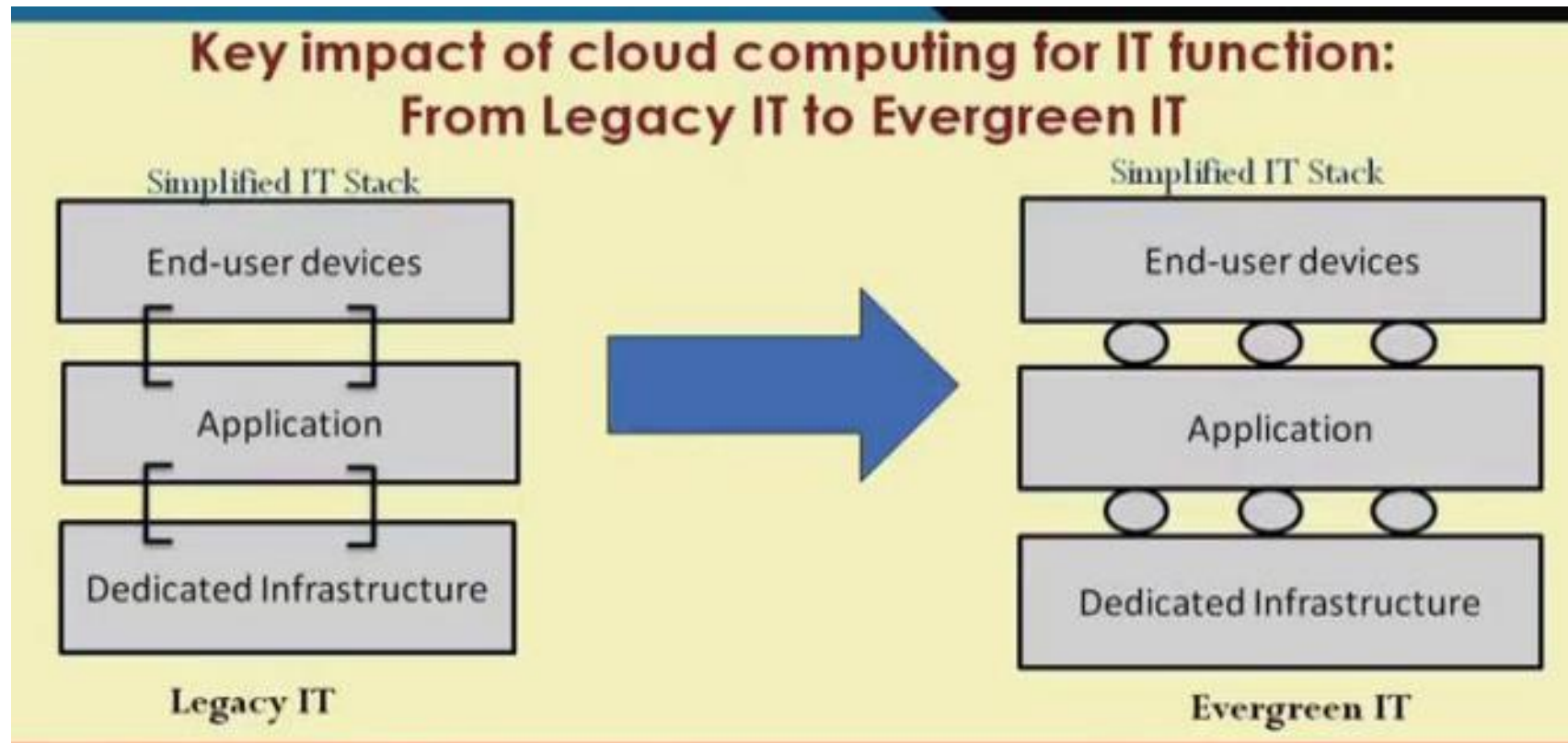
Traditional IT Approach



Google/Amazon
was made in a garage Which
means they had a server hosted in
their garage...

Example ?

Traditional IT Approach --> Simplified IT Approach



Example ?

Google

Corporate Office



Data Centre



1. Pay for Rent
2. Pay for maintenance
3. Pay for Damage etc.

Can you afford this ?

What do we want ?

01

We need "On-Demand" delivery of computation, memory, storage, database

02

We want to pay only for what we use - "Pay-as-you-go" pricing

03

Flexibility / Elasticity to expand and contract. Example: Electricity Supply in our house.

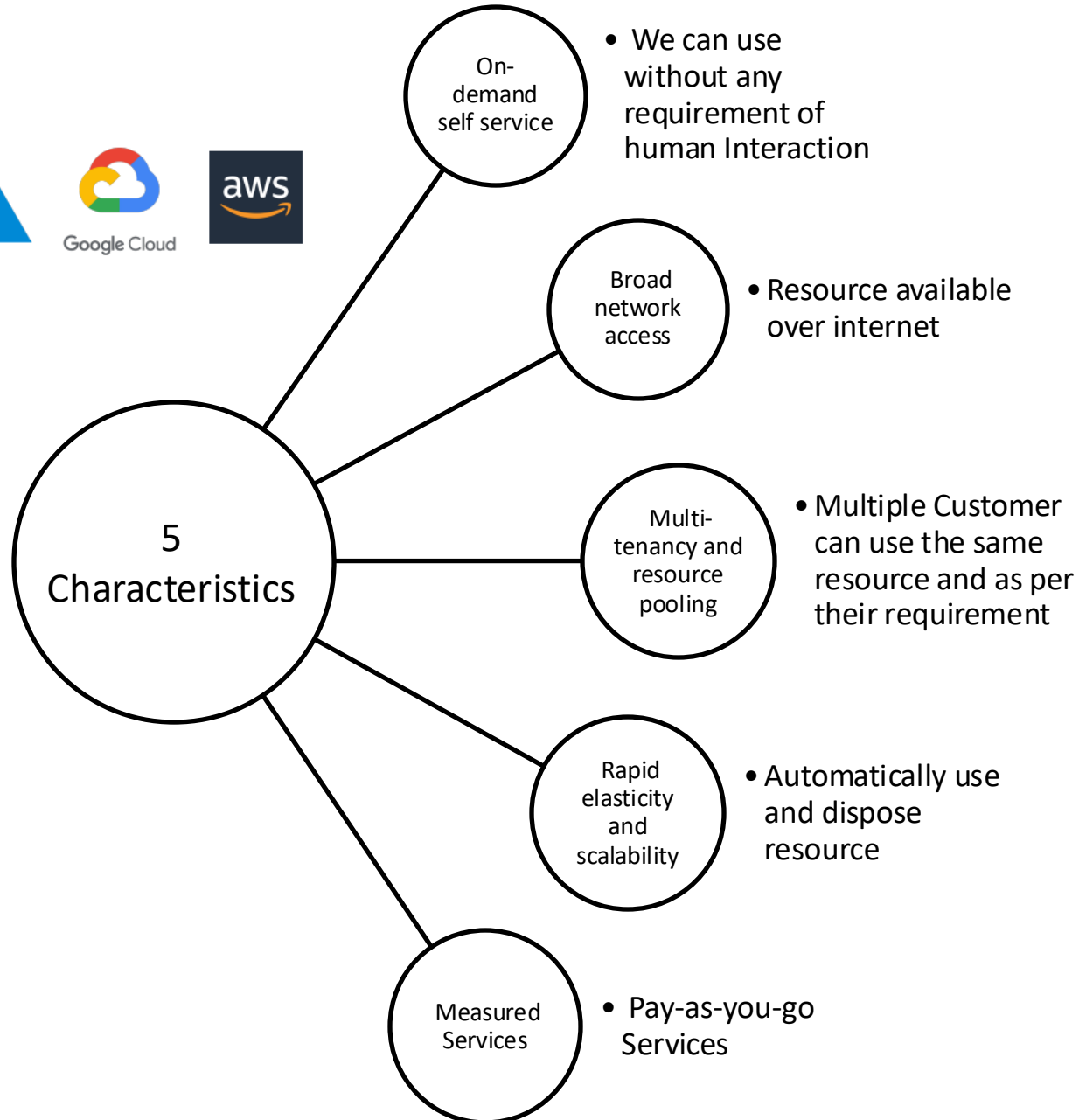
04

Easy-to-use UI for all type of users. Windows/Ubuntu etc.

05

Example : Google Drive, OneDrive, Gmail, Google Photos, Netflix....

Hmmm...but
what if I am a
institution?



Business-as-a-Service

Quality Attributes

Architectural Design

The Deployment Models of the Cloud

Private Cloud

- IITD--> Badal
- IITkgp-->Megh
- Complete Control, specific Business need, not accessible to public

Public Cloud

- Third Party Providers like AWS, Azure, Google Cloud etc.
-

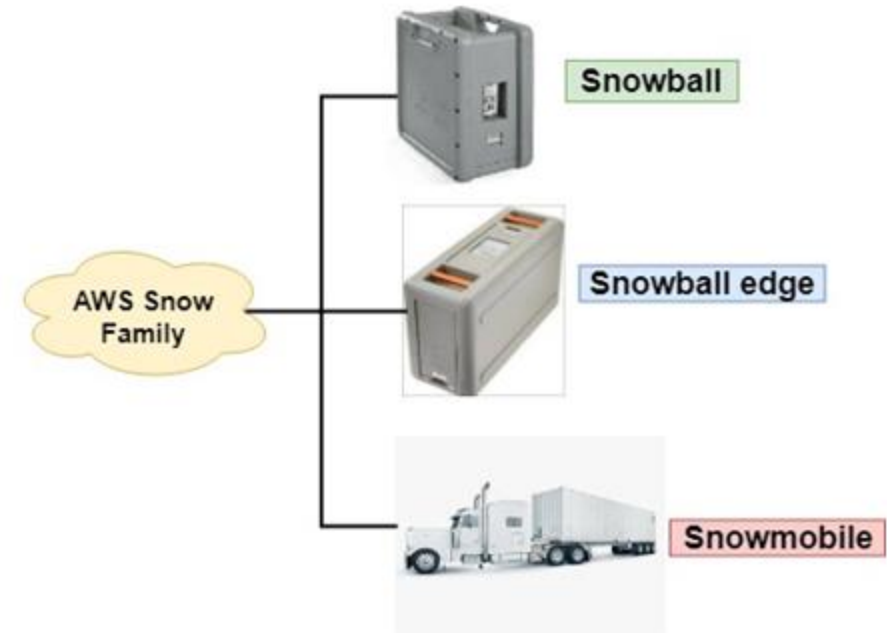
Hybrid Cloud

- Private+Public
- Keep some servers on premises and extend some capabilities to the Cloud

AWS Snow family-2016



Feature	AWS Snowball	AWS Snowmobile
Data transfer method	Physical device	Truck
Data transfer capacity	50 TB or 80 TB	100 PB
Transfer speed	Local network connection	up to 10 Gbps
Use case	Small-scale data migration	Extremely large-scale data migration



Sources:<https://www.geekwire.com/2016/use-amazons-snowball-snowballs-unleashes-45-foot-truck-model/>

Types of Cloud Computing

- **Infrastructure as a Service (IaaS) :** Infrastructure-as-a-Service (IaaS) is a form of cloud computing service that offers compute, storage and networking resources on-demand, usually on a pay-as-you-go basis. Businesses can purchase resources on-demand and as-needed instead of having to buy the hardware outright.

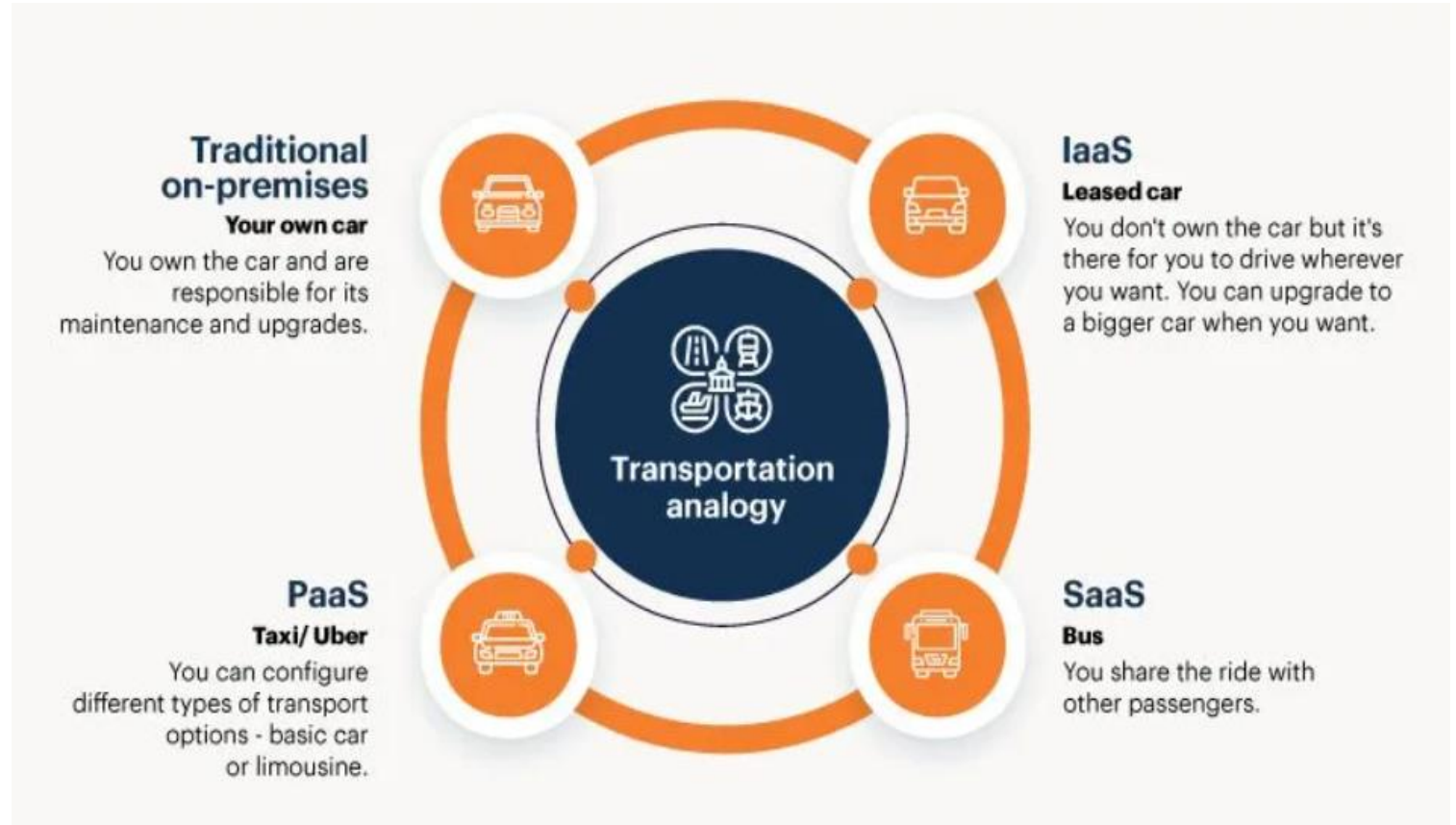
Amazon EC2 (on AWS)

- **Platform as a Service (PaaS) :** Platform-as-a-Service (PaaS) provides a framework for application creation and deployment. The technical stack (e.g., AWS Beanstalk, Google App Engine) required for application development is available on the cloud, which requires no download or local installation.

Windows Azure (Microsoft)

- **Software as a Service (SaaS) :** Software as a Service (SaaS) is a web-based software deployment model that makes the software accessible through a web browser. As a user of SaaS software, you don't care where the software is hosted, which operating system it uses, or which language it is written in.

Google Apps (Gmail)



Source: <https://www.eginnovations.com/blog/saas-vs-paas-vs-iaas-examples-differences-how-to-choose/>

Types of Cloud Computing

- **Infrastructure as a Service (IaaS)** : Infrastructure-as-a-Service (IaaS) is a form of cloud computing service that offers compute, storage and networking resources on-demand, usually on a pay-as-you-go basis. Businesses can purchase resources on-demand and as-needed instead of having to buy the hardware outright.

Amazon EC2 (on AWS)

- **Platform as a Service (PaaS)** : Platform-as-a-Service (PaaS) provides a framework for application creation and deployment. The technical stack (e.g., AWS Beanstalk, Google App Engine) required for application development is available on the cloud, which requires no download or local installation.

Windows Azure (Microsoft)

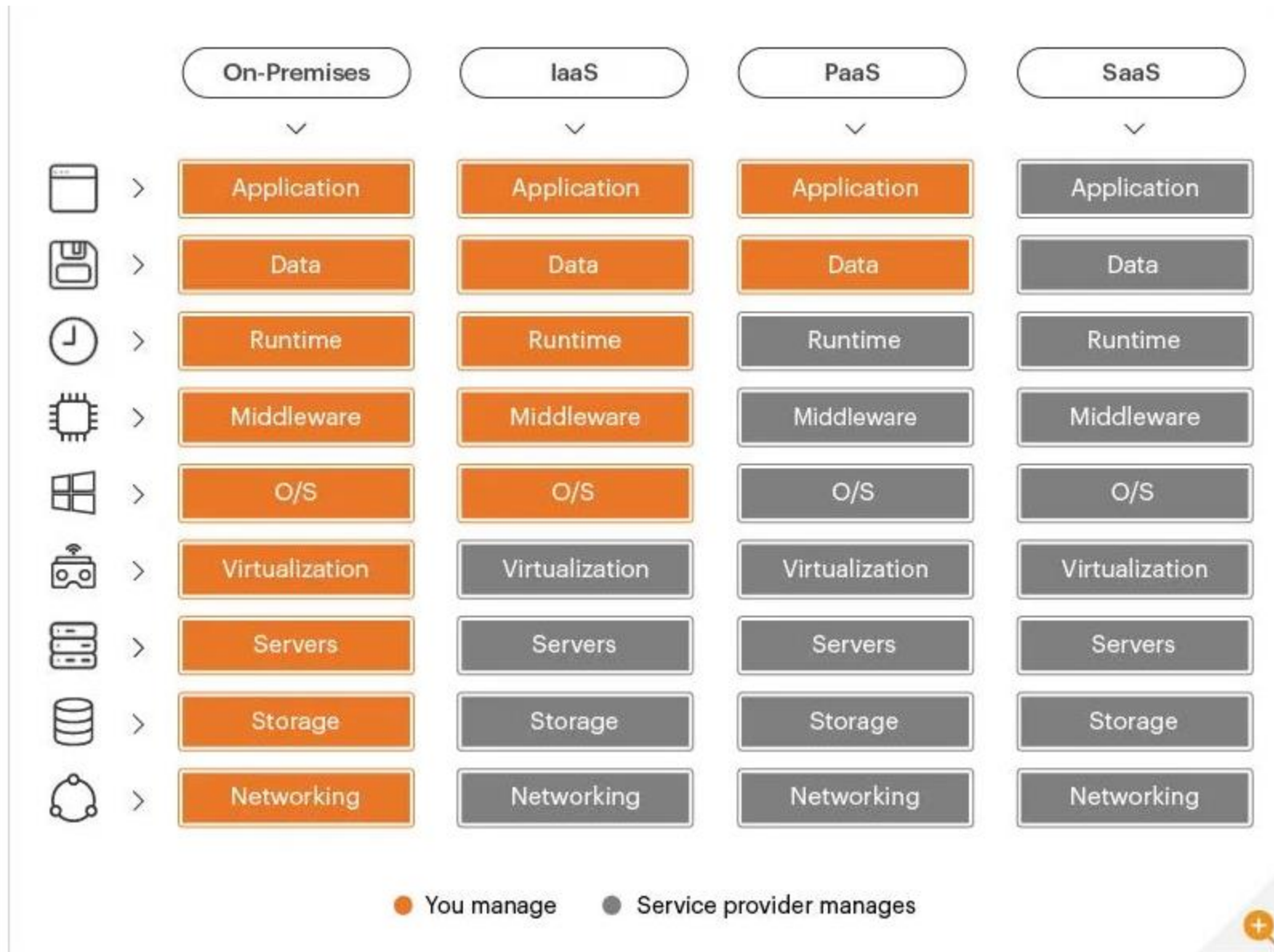
- **Software as a Service (SaaS)** : Software as a Service (SaaS) is a web-based software deployment model that makes the software accessible through a web browser. As a user of SaaS software, you don't care where the software is hosted, which operating system it uses, or which language it is written in.

Google Apps (Gmail)

Platform Type	Common Examples
SaaS	Google Workspace, Dropbox, Salesforce, Cisco WebEx, Concur, GoToMeeting
PaaS	Amazon Web Services (AWS) Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Red Hat OpenShift
IaaS	DigitalOcean, Linode, Rackspace, AWS, Cisco Metapod, Microsoft Azure, Google Compute Engine (GCE)

Source: <https://www.eginnovations.com/blog/saas-vs-paas-vs-iaas-examples-differences-how-to-choose/>

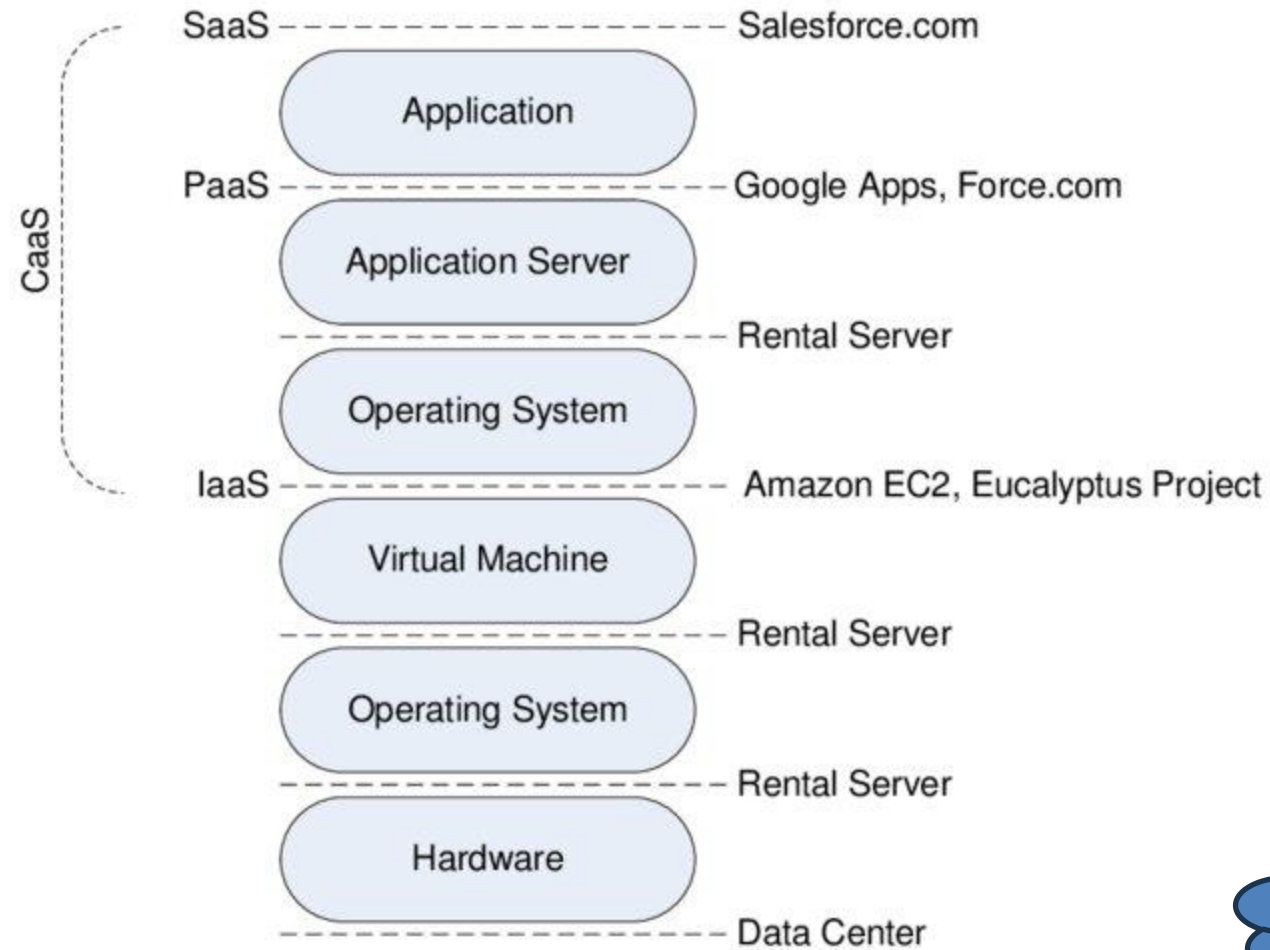
Stack Visualization



Advantages vs Disadvantages

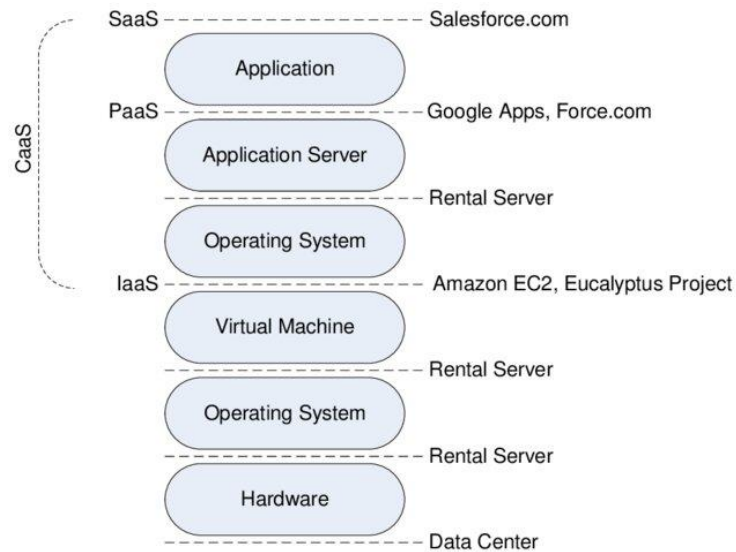
Type	Advantages	Disadvantages
SaaS	• Fast to realise benefits, quick set up and deployment • Lower up-front cost • Scalability and integration • New releases (upgrades) • Accessibility	• Lack of control • Vendor drives upgrades and new feature timelines • Security and data sovereignty concerns • Limited range of applications • Connectivity requirement • Performance concerns (speed, reliability, support)
PaaS	• Simple, cost-effective development and deployment of applications • Scalable • Highly available • No need for maintaining software • Significant reduction in the amount of coding required • Automation of business processes/policies	• Dependency on vendor • Security and data concerns • Compatibility of existing infrastructure • Performance concerns (speed, reliability, support)
IaaS	• Most flexible cloud computing model • Easy to automate deployment of storage, networking, servers, and processing power • Hardware purchases can be based on consumption • Clients retain complete control of their infrastructure • Resources can be purchased as needed • Highly scalable	• Security and data concerns • Legacy systems operating in the cloud • Internal resources and training • Multi-tenant security • Control over costs if scalability is unlimited

How to approach Cloud ?

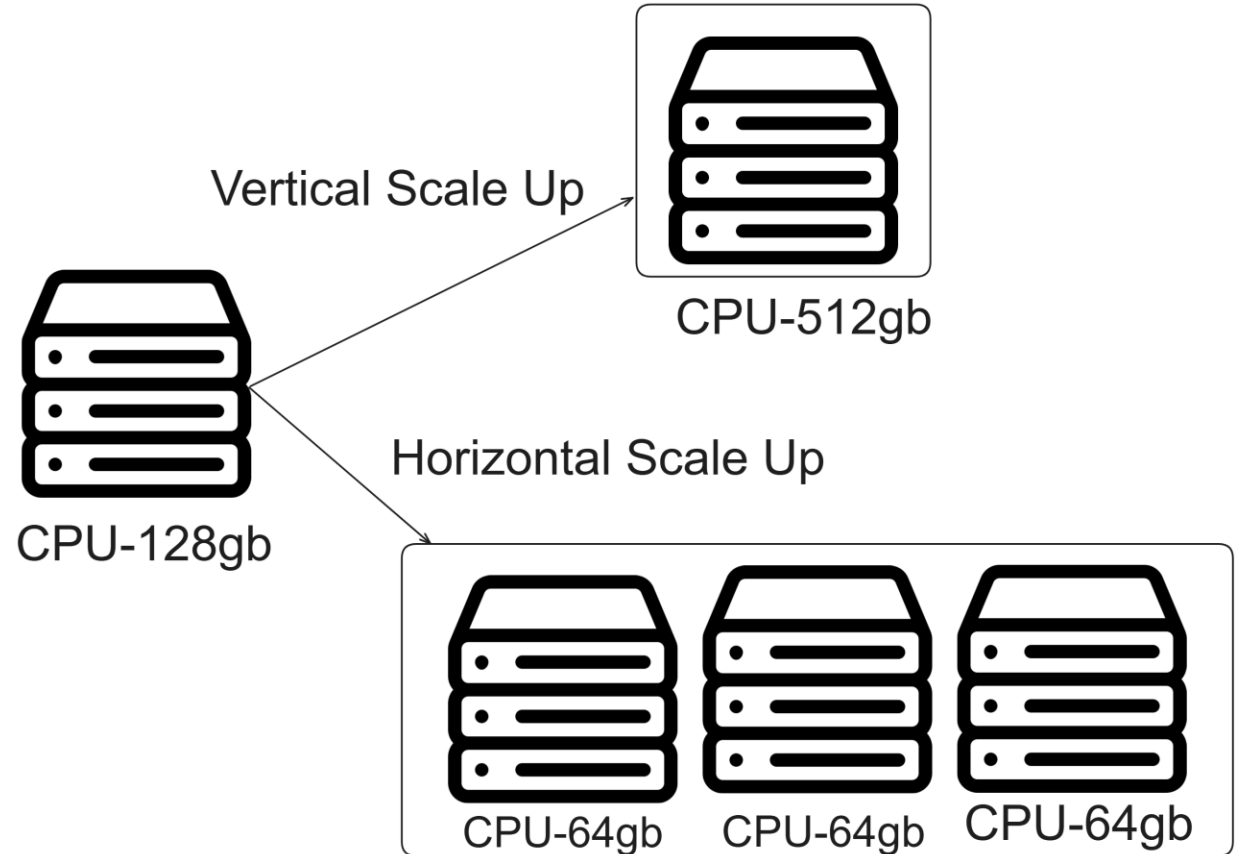


Based on your requirements...

How to approach Cloud ?



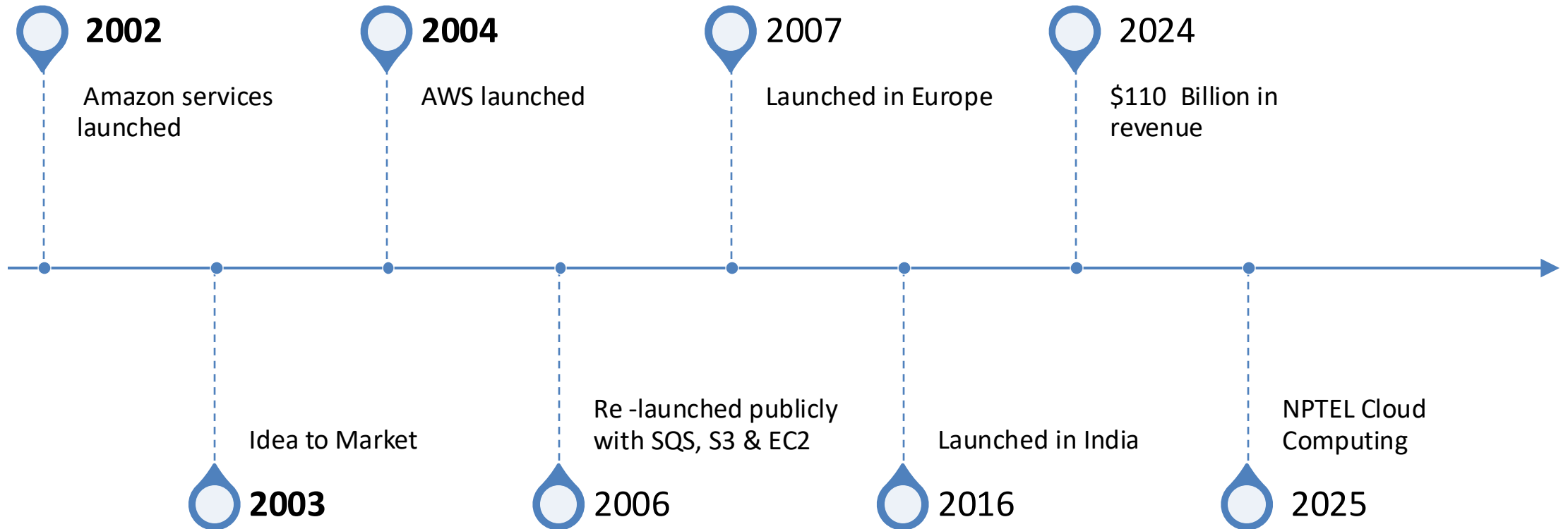
Based on your requirements...
1. Hardware



Hands-on Session

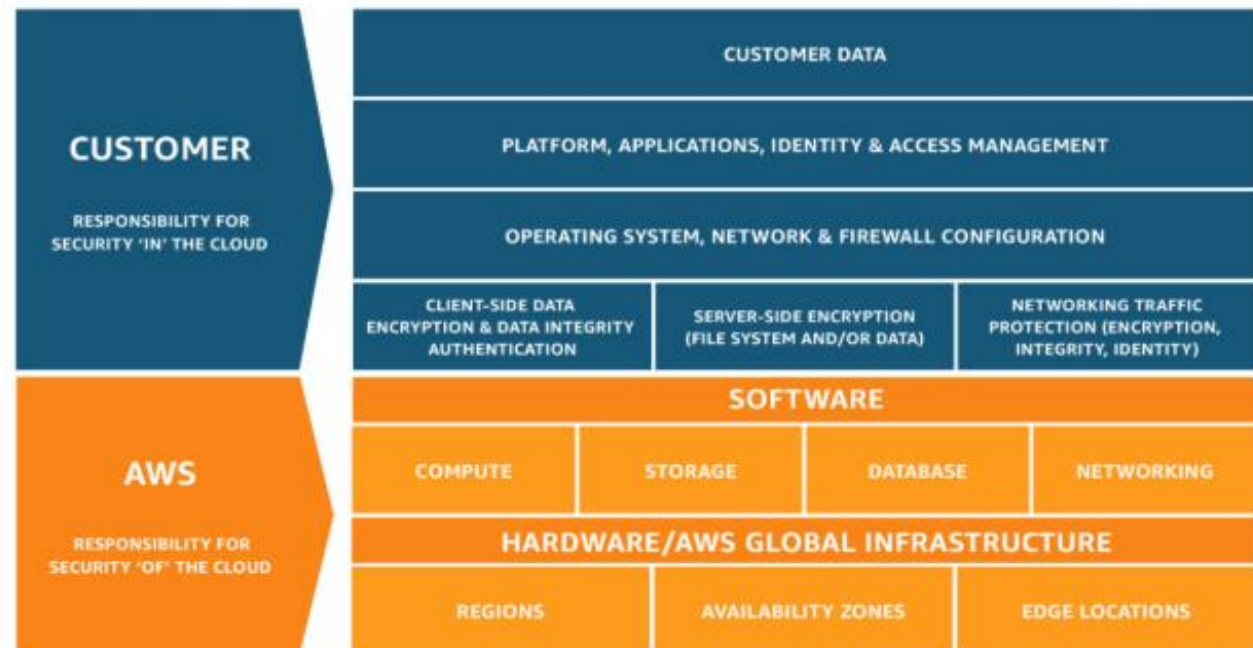
AWS

AWS Facts



AWS Important Points

- Map : https://aws.amazon.com/about-aws/global-infrastructure/regions_az/
- A region is a cluster of data centers • Most AWS services are region-scoped
- Each availability zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity
- **Account Opening : AWS Free tier account vs AWS Educate Account**
- **Shared Responsibility Model**



AWS Important Points

- Map : https://aws.amazon.com/about-aws/global-infrastructure/regions_az/
- A region is a cluster of data centers • Most AWS services are region-scoped
- Each availability zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity
- **Account Opening : AWS Free tier account vs AWS Educate Account**
- **Shared Responsibility Model**
- **NO illegal/ Harmful/ Scams can be done etc.**
- **Create Root Account with do email and phone verifications.**